QUALITY CONTROL FOR METAL CASTING PRODUCT

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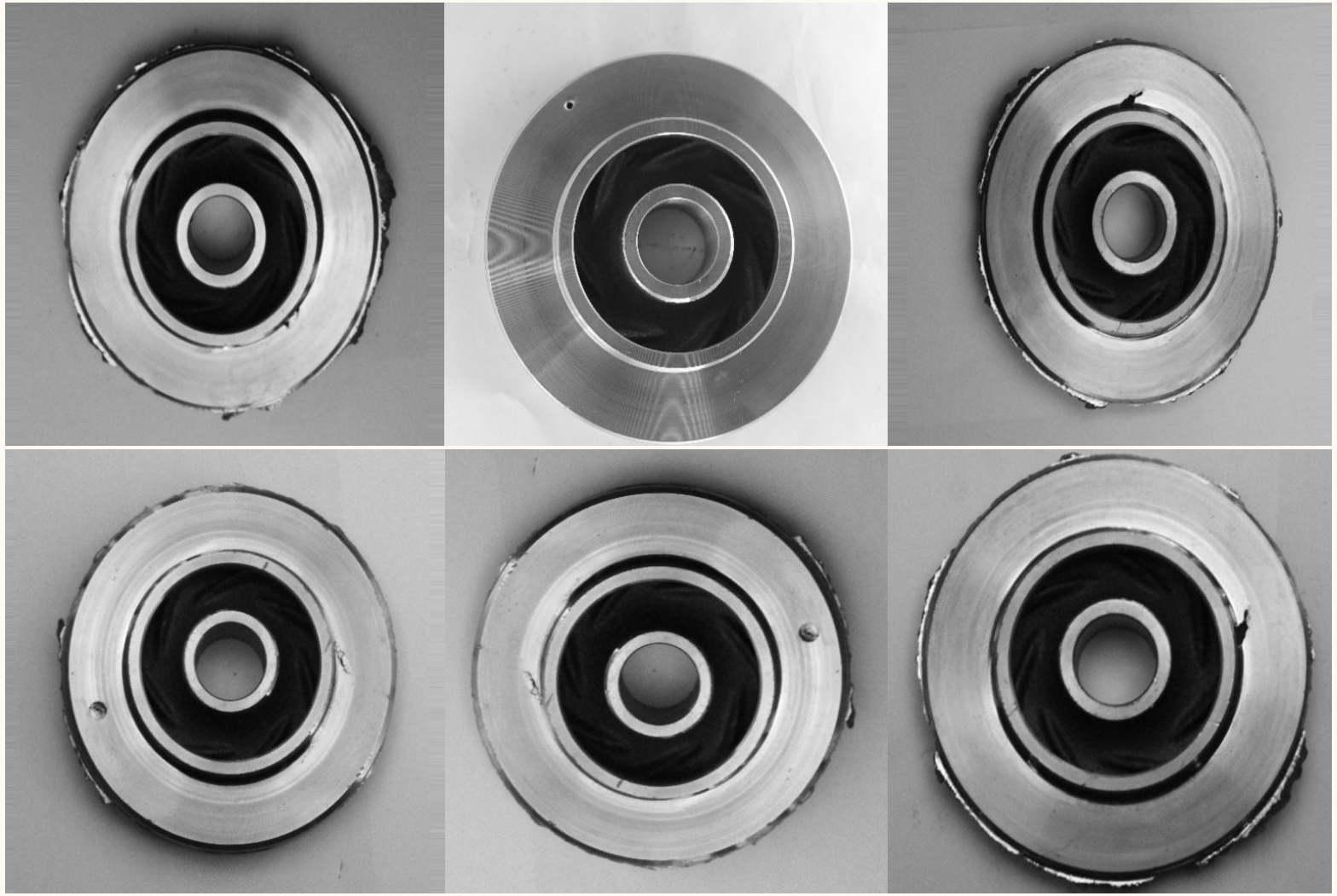
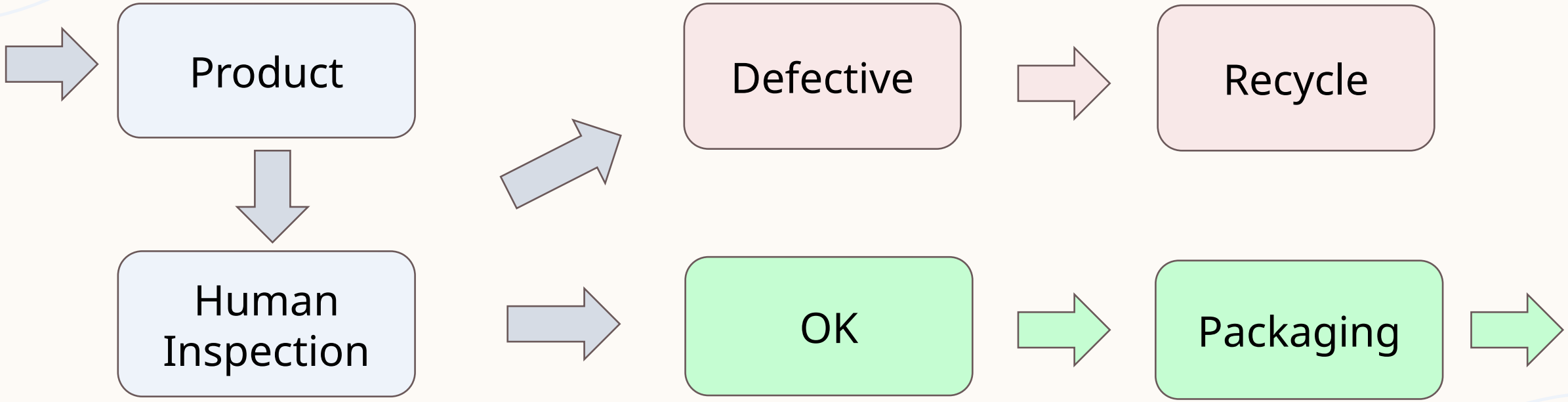
The Product

- **Name**: Submersible Pump Impe
- **Material**: Stainless Steel
- **Manufacturing Processing**: Cas
- **Types of Defective**: Blow holes, Pinholes, Burr, Shrinkage defects, Mold material defects, Pouring metal defects, Metallurgical defects

The Defective Products



Current Quality Inspection Process



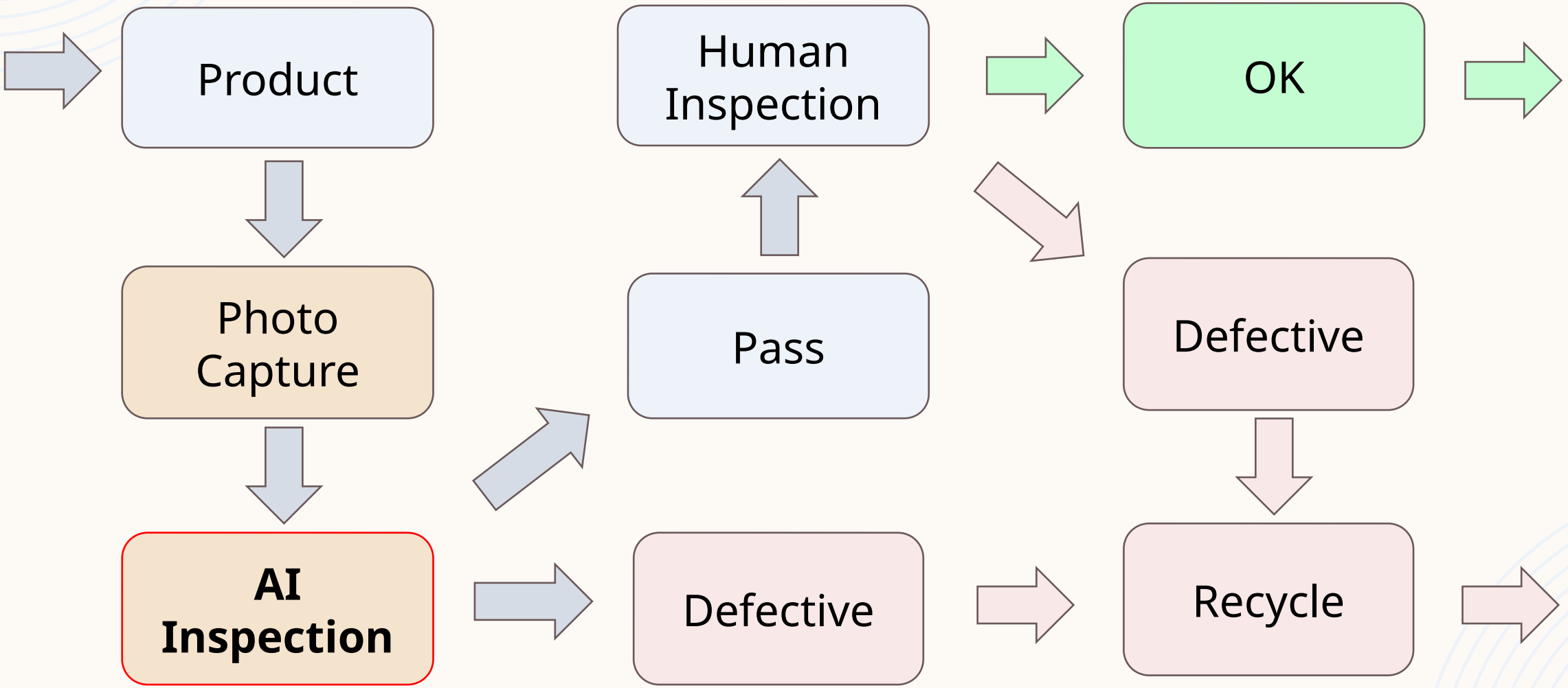
The Problem

Time-Intensive Inspections → **Delayed Order**

High Demand on Human Resources → **Increased Operational Costs**

Inconsistent Inspection Standards → **Potential Quality Issues**

The Solution



Step 1 : Set up Photo Capture Station

- **Continuous Lighting Source**:
  Utilization of dual Tungsten Light Bulbs for consistent and uniform illumination.

- **Temperature Regulation**:
  Deployment of an Infrared Thermometer to ensure product temperatures are maintained between 12°C and 35°C.

- **Efficient Product Handling**:
  Integration of a Conveyor Belt System for precise and automated transportation of products to the designated inspection area.

Step 2 : Implement AI-Powered Inspection

- **Developing the AI Model**:
  Training a Neural Networks model to integrate with Quality Control Software for advanced inspection capabilities.

- **Data Integration and Pipeline Creation**:
  Establishing a pipeline for seamless transfer and processing of image data into the neural network model.

- **Automated Product Handling and Sorting**:
  Utilizing AI model outputs to direct the automated transportation and sorting of products in the inspection area.

Step 3 : Human Inspection

- **Misclassification Analysis:**
  1. Inspect and identify misclassified products in both 'defective' and 'ok' categories.
  2. Separate non-defective items from the 'defective' area.
  3. Collect defective products from the 'ok' area for further analysis.
- **Model Improvement through Feedback Loop:**
  1. Reintegrate misclassified products into the training dataset.
  2. Retrain the model with the updated dataset to enhance accuracy and reduce future misclassifications.

Neural Networks Model Overview

- **Input**: 300*300 Greyscale Images
- **Output**: Binary Classification Signals
  - ‘ok’: Product passes the quality check
  - ‘defective’ : Product fails due to defects.
- **Accuracy: 80.30% (2023/11/16)**
- **Loss: 0.65**
- **Maintenance: Improve model with new image data**

Transforming Inspection

Time-intensive **Manual Operation**	→	**Fast, Automatic** **90% AI-Powered Inspection**
Heavy **Human Resource Cost**	→	**One-Time investigation with Minimal Maintenance**
Inconsistent **Inspection Standards**	→	**Continually Enhance Accuracy Over Time**

THANK YOU

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